AIRLINE PASSENGER SATISFACTION ANALYSIS

What turns an ordinary flight into a satisfying passenger experience?

PROJECT PROPOSAL • DESCRIPTIVE & DIAGNOSTIC ANALYTICS • NO MACHINE LEARNING

THE EXPERIENCE	THE SERVICE	THE DELAY
Which passenger groups feel most satisfied?	Which touchpoints may need attention?	How do delays relate to satisfaction?

01 | The Problem

A flight is more than getting from one place to another. A passenger's overall experience can be shaped by travel class, customer type, service quality and flight delays. Yet, without examining the data, it is difficult to know which patterns deserve attention.

Core question: What does the passenger data reveal about satisfaction—and where could the airline improve?

02 | Project Objectives

- Measure overall satisfaction and compare results across passenger groups.
- Identify service attributes with comparatively low ratings.
- Explore how satisfaction varies by travel class, customer type and travel purpose.
- Examine departure and arrival delay patterns alongside satisfaction levels.
- Turn the findings into clear, evidence-based recommendations.

03 | About the Dataset

The project will use the Airline Passenger Satisfaction dataset available on Kaggle. The commonly used version contains approximately 129,880 records, with fields covering passenger and travel details, service ratings, flight delays and satisfaction category. The exact row count and columns will be confirmed after download.

Dataset source: kaggle.com/datasets/teejmahal20/airline-passenger-satisfaction

04 | Tools & Approach

PYTHON	POWER BI
Inspect and clean data	Create DAX calculations
Explore patterns	Build interactive dashboard

The work will focus on descriptive and diagnostic analytics only. No Machine Learning or predictive modeling will be used.

05 | Insights We Aim to Discover

- What proportion of passengers are satisfied, neutral or dissatisfied?
- Which passenger segments show relatively higher or lower satisfaction?
- Which service ratings are weakest and may warrant review?
- How do delay patterns differ across groups, and how do they align with satisfaction?

Important: These are questions to investigate, not pre-decided results. Final insights will be based on the analysis.

06 | Expected Conclusion

The project is expected to produce a clear picture of satisfaction patterns, service areas that may need attention, and passenger groups with different experiences. The final recommendations will be supported by Python analysis and a Power BI dashboard. Conclusions will describe observed relationships without claiming that one factor necessarily causes another.

PROPOSED OUTCOME

A concise analytical report and an interactive Power BI dashboard that help communicate the most useful findings and practical service-improvement opportunities.

Submitted for trainer review and project approval